L08: Little's Law

Simulation Without a Black Box

Outline

Learning Objectives

By the end of this lecture you will be able to:

- State Little's Law and its conditions for applicability.
- Explain the sample-path proof using the area-under-curve argument.
- Use the decomposition $L = L_q + \rho$ to relate system and queue quantities.
- Verify Little's Law numerically from the HW-02 trace.
- Solve for any one of L, λ, W given the other two.
- Articulate what Little's Law does *not* tell you (the distribution of W).

Little's Law: Statement

Little's Law (J.D.C. Little, 1961)

For any stable system in steady state with a well-defined boundary:

$$\boxed{L = \lambda W}$$

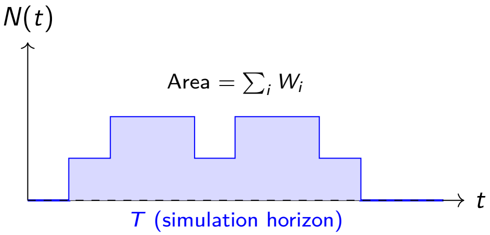
and equivalently for the queue subsystem:

$$L_q = \lambda W_q$$

where λ is the long-run average arrival rate ($=$ throughput rate in steady state), W is the average time an entity spends in the system, and L is the average number of entities in the system.

- Requires: stable system ($\lambda <$ service capacity), well-defined boundary.
- Does **not** require: Poisson arrivals, exponential service, single server, FCFS.
- Holds for any distribution, any number of servers, any discipline.

Sample-Path Proof: Area Under $N(t)$



Key identity:

$$\int_0^T N(t)\,dt = \sum_{i=1}^{n} W_i$$

Each entity's sojourn time W_i contributes exactly W_i units of area.

Divide both sides by T:

$$\underbrace{\frac{1}{T}\int_0^T N(t)\,dt}_{L} = \underbrace{\frac{n}{T}}_{\lambda} \cdot \underbrace{\frac{1}{n}\sum_{i} W_i}_{W}$$

Therefore $L = \lambda W$. $\square$

Decomposition: $L = L_q + \rho$

Split the system into the *queue* and the *server*:

$$L = L_q + L_s$$

For the server subsystem: $L_s = \lambda W_s = \lambda \cdot (1/\mu) = \lambda/\mu = \rho$.

Decomposition formula

$$L = L_q + \rho, \qquad W = W_q + \frac{1}{\mu}$$

Practical use:

- If you know L_q and ρ, you can compute L without re-running the simulation.
- $W = W_q + \bar{S}$ where $\bar{S}$ is mean service time — exact even for complex service distributions.
- Verifying $L \approx L_q + \rho$ is a quick sanity check on simulation output.

Numerical Verification from HW-02 Trace

HW-02 trace results (10 customers, $T = 60$ min, $\mu = 1/\bar{S}$):

Quantity	From trace	Little's Law check
n	10	—
T	60 min	—
$\hat{\lambda}$	$10/60 = 0.167/\text{min}$	—
$\bar{W}$	(from trace)	—
$\bar{L}$	(area/T)	$\approx \hat{\lambda}\bar{W}$?
$\bar{W}_q$	(from trace)	—
$\bar{L}_q$	(area/T)	$\approx \hat{\lambda}\bar{W}_q$?
$\hat{\rho}$	(busy-time/T)	$\approx \bar{L} - \bar{L}_q$?

Fill in the blanks from your completed HW-02 trace. Both checks should hold to two decimal places (small discrepancy is rounding, not an error in Little's Law).

Applications: Solve for the Unknown

Given any two of $\{L, \lambda, W\}$, compute the third:

Known	**Unknown**	**Formula**	**Example**
λ, W	L	$L = \lambda W$	$L = 2/\text{hr} \times 3\text{hr} = 6$
L, W	λ	$\lambda = L/W$	$\lambda = 6/3 = 2/\text{hr}$
L, λ	W	$W = L/\lambda$	$W = 6/2 = 3\,\text{hr}$
λ, W_q	L_q	$L_q = \lambda W_q$	Emergency dept. example
L_q, λ	W_q	$W_q = L_q/\lambda$	Throughput accounting

Practical scenario: A hospital manager observes an average of 12 patients in the ED and knows the arrival rate is $4/\text{hr}$. Without simulation, Little's Law gives $W = 12/4 = 3\,\text{hr}$.

What Little's Law Does *Not* Tell You

Little's Law is about means only

It relates the *mean* number in system to the *mean* throughput rate and the *mean* sojourn time. It says nothing about:

- The **distribution** of W (you cannot infer variance, skewness, or percentiles).
- The **distribution** of $N(t)$ (you cannot infer the probability of overflow).
- **Transient** behavior (Little's Law is a steady-state result; it does not apply during warm-up).
- **Which** customers wait and for how long (no individual-level information).
- Performance under **priority disciplines** (different classes have different W; use separately per class).

For the full distribution of W, you need simulation output — not just the mean.

Activity: Verify $L = \lambda W$ for 200-Customer M/M/1

Parameters: $\lambda = 3/\text{hr}$, $\mu = 4/\text{hr}$, $\rho = 0.75$.

Analytical predictions:

$$W = \frac{1}{\mu - \lambda} = \frac{1}{1} = 1\,\text{hr}, \qquad L = \lambda W = 3 \times 1 = 3, \qquad L_q = \rho^2/(1-\rho) = 2.25.$$

Your tasks:

1. Run `trace_single_server()` (or SimPy M/M/1) for 200 customers.
2. Compute $\hat{\lambda}$, $\bar{W}$, $\bar{L}$, $\bar{L}_q$, $\hat{\rho}$ from output.
3. Check: $\bar{L} \approx \hat{\lambda}\bar{W}$? How close?
4. Check: $\bar{L} \approx \bar{L}_q + \hat{\rho}$?
5. Compare all five quantities to the M/M/1 formulas. Discuss discrepancy.

Expected: Little's Law holds to within 5% at $n = 200$. The M/M/1 formula may differ more because of finite-sample and transient effects.

Key Takeaways

- Little's Law $L = \lambda W$ is an exact result for any stable system with a well-defined boundary.
- The proof is a pure sample-path argument: area under $N(t)$ equals sum of sojourn times.
- Decomposition $L = L_q + \rho$ links system, queue, and server quantities.
- Use Little's Law to cross-check simulation output and to compute one unknown from two knowns.
- Little's Law says nothing about the *distribution* of W — for that, run the simulation and collect all W_i.
- In steady state, L, W, and λ form a triangle: know any two, compute the third.

Next module: SimPy basics and the process-oriented worldview (M04, L09).